

inPowerSuite E-Commerce APIs

Niche Applications, Inc.

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APPENDIX

Appendix A - Parameters

ERROR! BOOKMARK NOT DEFINED.

Error! Bookmark not defined.

Overview

inPowerSuite contains a set of programming functions or application programming interfaces (APIs) that can be used for integrating e-commerce functionality into a custom-built web store. This API set allows a **web** development team to leverage *inPowerSuite*'s native functionality for managing Customers, Shopping, Secure Ordering and Tracking capabilities, and combine it with an existing website.

This API provides database and e-commerce insulation when developing a web store using ODBC, PL/SQL, ASP, C++, ColdFusion, and Perl programming languages. Equally as important, it is useful in those situations that necessitate shielding developers from both the complexities of a relational database and the SQL programming of the required e-commerce transactions.

The *inPowerSuite* API is a fundamental building block for any company that endeavors to create a robust e-commerce web store while exercising their own control over the store's overall presentation and shopping experience.

API Function Groups

inPowerSuite provides a rich set of APIs for building an e-commerce web store. These numerous API functions are organized into five logical groups, thus making it effortless for a web development team to assimilate the complexity of e-commerce. The groups are:

General API Functions

This group of API functions manages the tasks for initializing a shopping session; handling errors; retrieving general store settings, and sales representatives information.

Shopping API Functions

This group of API functions manages the tasks for retrieving content for the shopping catalog, primarily consisting of: categories, product lists, product details, searching, and suggestive selling arrangements for a product.

Cart API Functions

This group of API functions manages the tasks for manipulating a shopping cart, primarily consisting of: adding and removing items; and changing item quantities.

Checkout API Functions

This group of API functions manages the tasks for completing a web order, primarily consisting of: authenticating a returning customer, registering a new customer, gathering shipping information, obtaining freight charges, calculating taxes, processing a credit card transaction, and finally providing confirmation of an order.

OrderTracking API Functions

This group of API functions manages the tasks for obtaining tracking information for an existing order.

Calling Method

return_code = NGSAPI.function (IN parameter1, 2, ..., OUT parameter1, 2, ...) where,

return_code

A local integer variable defined in the program that calls the inPowerSuite API function. This variable will contain a value of zero (0) or one (1) after execution, to represent either success or failure, respectively. On failure, a subsequent call should be made to the **get_error_list()** API function to obtain error information.

Additionally, any API function call that returns no result or the "Not found" condition will have a return_code value set to one (1). Subsequently, the same API function will generate error information that can be obtained using **get_error_list()**.

function

The name of the *inPowerSuite* API function used in the NGSAPI library.

IN parameter1, 2, ...

Input parameters having data types as defined below. All **IN** parameters are declared **first** in the formal parameter list.

OUT parameter1, 2, ...

Output parameters are placeholders for results being returned from the *inPowerSuite* API function call and have data types as define below. All **OUT** parameters are declared **last** in the formal parameter list.

Parameter Data Types

The API functions require parameters that are defined as primitive data types (to support open application interoperability). The data types are:

Number

Any signed binary or scalar type that contains numeric, rate or currency information.

String

Any ASCII, text, or alphanumeric character information.

Date String

Any valid date or date/time value formatted as "mm/dd/yyyy" (short date format) or "mm/dd/yyyy hh24:mi:ss" (long date format), respectively.

inPowerSuite APIs will accept either format for an **IN** parameter, but will always return the long date format as an **OUT** parameter.

Ref Cursor

A PL/SQL data type that returns a result set (a.k.a, set of records).

Conventions

- ☐ All API functions are located in the PL/SQL package named NGSAPI.
- ☐ All **IN** and **OUT** parameters must be declared and passed for each API call.
- ☐ A calling program must call the appropriate **init_session()** API function prior to any other API.
- ☐ The *inPowerSuite* API functions must be called from a "state-full" program.
- ☐ Handling the NULL condition. The API will accept a non-NULL value that represents NULL for each parameter data type as follows:
 - String A pair of single quotes, representing an empty string ''.
 - Number
 - Date
- ☐ Transaction Management for database commits and rollbacks

API Summary

General API Functions

init_session (IN fv_organization_code, fv_source_code, OUT fn_session_id)

Returns a unique session ID that calling program can use to maintain state. This is the first function that should be called by an external program using the *inPowerSuite* API. Its primary purpose is to set the context for the organization whose web store data will be manipulated (i.e. *inPowerSuite* supports multiple organizations in one installation).

InPowerSuite will create a unique session ID for each session and returns this ID as an output parameter for use in the calling program, if so desired.

init_session (IN fv_organization_code, fv_source_code, fn_customer_id, OUT fn_session_id)

This is an overloaded function call that does the same thing as the previous version and additionally will set the context of the customer ID, if known. This will allow a specific customer price list to be enabled for the shopping session.

get_store_parameter (IN fv_parameter_name, OUT fc_web_store_parameter)

Returns the parameter definition for the specified web store parameter. This API function should be used to obtain those settings that are pertinent to the store's overall presentation. A full description of each parameter can be obtained from the *inPowerSuite Parameters* document.

get_error_list (OUT fc_error_list)

Returns a list of errors when error code is one (1), from the previous API call. The ref cursor contains the following information.

get_sales_person_list (OUT fc_sales_person_list)

Returns a list of sales persons (representatives) for the web store.

Shopping API Functions

get_plugin_list (OUT fc_category_list)

Returns a result set of active Plug-ins available on the web store, using the same ref cursor as defined for **get_category_list()** API function.

get_category_list (IN fn_start_category_id, fn_max_level, OUT fc_category_list)

Returns a tree (hierarchy) of active categories starting at the specified category, up to the max level. If the starting category is zero (0), then all categories are returned up to the max level. If max level is zero (0) then all active categories are returned.

get_category_id (IN fv_category_name, OUT fn_category_id)

Returns the ID for the specified category. This API function should be used before obtaining more category information from a category name.

get_category_product_list (IN fn_category_id, OUT fc_product_list)

Returns a result set of products for a specified category. The ref cursor contains the following information for products and/or product SCS groups.

get_related_product_list (IN fn_product_id, fv_relationship_type, OUT fc_product_list)

Returns a result set of products that are related to the specified product and its relationship type, using the same ref cursor as defined for **get_category_product_list()** API function.

get_product_list (IN search_pattern, search_category, OUT fc_product_list)

Returns a result set of products and/or product SCS groups based on the specified Search Pattern and Category. The search is performed across all search-indexed columns in the database. If Category is not specified then all products are searched.

get_product_info (IN fn_product_id, OUT fc_product_info_list)

Returns the details for the specified product. This information can be displayed in any arrangement that is deemed appropriate for your web store.

Cart API Functions

init_cart (OUT fn_order_id)

Returns a new order ID after initializing an order for the current shopping session.

Note: An *inPowerSuite* order record is initialized with a status of "new" to represent a **cart** and not a true order. Subsequently, the order record then becomes a true order only when the order status is promoted to "booked", thus after the order payment has been authorized.

The calling program should obtain an order ID with this API before calling the **create_order_line()** API function to place a product in the shopping cart.

add_cart_item (IN fn_order_id, fn_product_id, fn_product_quantity, OUT fn_order_line_id)

This is an overloaded function that returns a new order line ID after adding the specified product to the shopping cart.

add_cart_item (IN fn_order_id, fv_product_number, fn_product_quantity, OUT fn_order_line_id)

This is an overloaded function that returns a new order line ID after adding the specified product to the shopping cart.

get_cart_items (IN fn_order_id, OUT fc_cart_item_list)

Returns a list of products currently in the shopping cart for the specified order.

change_cart_item_quantity (IN fn_order_line_id, fn_new_quantity)

Updates the quantity for the specified product in the shopping cart. If the new quantity is zero (0) then the item is removed from the shopping cart.

remove_cart_item (IN fn_order_line_id)

Removes the specified product from the shopping cart, using the **change_cart_item_quantity()** API setting fn_new_quantity IN parameter to zero (0).

Note: This function will physically delete an order line from a shopping cart and not a "live" order.

Checkout API Functions

init_customer (IN fn_customer_id)

Initializes the global private variable gn_cust_id. It is not necessary for the calling program to use this API when creating a new customer, as the **create_customer()** API function makes an internal call to it.

add_customer (IN fn_session_id, ..., OUT fn_customer_id)

Returns a new customer ID after creating a new customer with billing information, and calling **init_customer()** API function to set the global private variable gn_customer_id.

authenticate_customer (IN fv_username, fv_password, OUT fn_customer_id)

Returns an existing customer ID after authenticating the returning customer based on the specified login credentials. It subsequently calls **init_customer()** API function to initialize the global private variable gn_cust_id.

The calling program must use this API function before any customer centric functions can be performed on an existing customer.

verify_customer (IN fv_email_address)

Verify a customer who has forgotten his/her password, based on valid a email address (that exists on file.)

get_customer_billing_info (IN fn_customer_id, OUT fc_customer_billing_info)

Returns the billing information for the specified customer.

get_customer_shipto_list (IN fn_customer_id, OUT fc_customer_shipto_list)

Returns a list of active shipping addresses for the specified customer.

add_customer_shipto (IN fn_customer_id, fv_shipto_first_name,..., OUT fv_shipto_address_id)

Returns a new shipping address ID after adding the address for the specified customer.

change_customer_shipto (IN fn_customer_id, fv_shipto_address_id,...)

Saves changes for an existing customer's shipping address.

Note: Setting fn_shipto_inactive_flag to one (1) instructs *inPowerSuite* to perform a logical delete of the shipping address, thus disabling it from future use.

Checkout API Functions (con't)

set_order_line_shipto (IN fn_order_line_id, OUT fn_shipto_address_id)

Sets the shipping destination for the specified order line, as *inPowerSuite* natively supports split shipments at the order line level.

set_order_shipto_charge (IN fn_order_id, fv_ship_method_code, fn_amount)

Sets the selected shipping method and charge for the specified order.
This API function should be used only when one (1) shipping destination exists for the order.
Otherwise, the calling program must use the **set_order_line_ship_charge()** API function instead.

set_order_line_shipto_charge (IN fn_shipto_address_id, fv_ship_method_code, fn_amount)

Sets the selected shipping method and charge for the specified shipping destination on the order.
This API function should be used only when multiple shipping destinations exist for the order.
Otherwise, the calling program must use the **set_order_ship_charge()** API function instead.

get_payment_methods (OUT fc_payment_method_list)

Returns a list of payment methods that have been defined for the web store.

validate_card (fv_card_type, fv_account_number)

Validates a credit card, consisting of the LUHN MOD-10 check, and the validity of the card type.

authorize_payment (IN fn_order_id, fv_payment_type, fv_account_number, fv_expiration_date)

Validates, and authorizes a payment with the credit card processor for the specified order.
inPowerSuite automatically generates an order confirmation email on successful application of the payment.

Order Tracking API Functions

get_customer_order_list (IN fn_customer_id, OUT fc_customer_order_list)

Returns a list of orders for the specified customer.

get_order_info (IN fn_order_id, OUT fc_order_header_info, fc_order_totals)

Returns confirmation information for the specified order. This API is generally needed after processing credit card payment for a new order or for obtaining information for order tracking purposes.

The results from this API function is described in **get_order_header_info()** and **get_order_totals()** API functions.

get_order_header_info (IN fn_order_id, OUT fc_order_header_info)

Returns header information for the specified order.

get_order_shipto_info (IN fn_order_id, OUT fc_order_shipto_info)

Returns sub-totals for each ship-to address for the specified order.

.....

get_order_lines (IN fn_order_id, OUT fc_order_line_list)

Returns list of order lines for the specified order including product, quantity, price, sales tax, destination and related charges for the destination.

.....

get_order_shipto_tracking (fn_order_SH_charge_id, OUT fc_order_shipto_tracking)

Returns list of tracking numbers and possibly, their related URLs to obtain tracking details for the specified order-shipping destination. This API function should be called after using the **get_order_shipto_info()** API function to obtain a list of shipping destinations for an order.

.....

get_order_totals (IN fn_order_id, OUT fc_order_totals)

Returns the grand totals (including Sales Tax and Shipping & Handling charges) for the specified order. One row is returned for a valid order ID.

.....

get_order_payment_list (IN fn_order_id, OUT fc_order_payment_list)

Returns payment information for the specified order. This includes all payment adjustments made to the specified order, in event of product returns and/or cancellations.

.....

General API Functions

init_session (IN fn_session_id , fv_organization_code, fv_source_code, OUT fn_session_id)

Returns a session ID that can be used by the calling program to maintain state. This is the first function that should be called by an external program using the inPowerSuite API.

The primary purpose of this API function is to set the context for the organization (legal entity) whose web store data will be manipulated for either a brand new session, or an existing session. (i.e. inPowerSuite supports multiple organizations in one installation).

inPowerSuite will create a session ID when this API function is called for the very first time to establish a new shopping session, if an existing session ID (value) was not passed in. Otherwise, inPowerSuite validates the received session ID and set the context for the current shopping session.

IN – PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_session_id	Number		Represents the unique number (previously generated by the database) to identify a shopping session. Note: If the value is Blank, then this API function will generate a new session ID, otherwise it will set the context for the existing session.
2.	fv_organization_code	String	3..12	Represents the organization code that has been defined for the web store.
3.	fv_source_code	String	3..15	Represents the name of the source application, which is calling this inPowerSuite API function. This code is primarily used for auditing and tracing purposes.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_session_id	Number		Represents the unique number generated by the database to identify a shopping session.

Note: The init_session API function is the precursor for any other API function call. In other words, every inPowerSuite API function call must be preceded by this API function, which is used establish the session state for the impacted API function.

init_session (*IN fn_session_id, fv_organization_code, fv_source_code, fn_customer_id, OUT fn_session_id*)

This is an overloaded function call that does the same thing as the previous version and additionally will set the context of the customer ID, if known. This will allow a specific customer price list to be enabled for the shopping session.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_session_id	Number		Represents the unique number (previously generated by the database) to identify a shopping session. Note: If the value is Blank, then this API function will generate a new session ID, otherwise it will set the context for the existing session.
2.	fv_organization_code	String	3..12	Represents the organization code that has been defined for the web store.
3.	fv_source_code	String	3..15	Represents the name of the source application, which is calling this inPowerSuite API function. This code is primarily used for auditing and tracing purposes.
4.	fn_customer_id	Number		Represents the customer ID that was obtained from a cookie setting.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_session_id	Number		Represents the unique number generated by the database to identify a shopping session.

get_store_parameter (*IN fv_parameter_name, OUT fc_web_store_parameter*)

Returns the parameter definition for the specified web store parameter. This API function should be used to obtain those settings that are pertinent to the store's overall presentation. A full description of each parameter can be obtained from the **inPowerSuite Parameters** document.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fv_parameter_name	String		Represents the name of the specified web store parameter.

FC WEB STORE PARAMETER

Seq.	Column	Data Type	Data Length	Description
1.	Parameter_Name	String		Represents the name of the specified web store parameter.
2.	Parameter_Type	String		Represents the data type of the parameter. Values are: T Text or String D Date/time N Numeric F Flag for On/Off, Yes/No, True/False
3.	Text_Value	String		Contains the parameter value when the parameter_type is TEXT.
4.	Number_Value	Number		Contains the parameter value when the parameter_type is NUMBER.
5.	Date_Value	Date Time String		Contains the parameter value when the parameter_type is DATE.
6.	Flag_Value	String	1	Contains the parameter value when the parameter_type is FLAG.

get_error_list (OUT fv_error_list)

Returns a list of errors (for the shopper) when the return code is one (1), for the previous API function call.

The error list contains at least one error or a group of errors within a single text string where each error is separated by a carriage return character.

Each error is further composed of two sub-strings that are separated by a colon and a space (": ") characters. The first part of the string represents the web store's error code (i.e. **NGS-nnn**), while the latter part of the string represents the error text (i.e. **"Customer Last Name is missing."**)

Best Practices

1. The calling program should always expose the error information (obtained from the error list) to the shopper.
2. The calling program should provide a standard error title when displaying the error information to the shopper. (i.e. **"Error!"**)
3. In the event that the web server is unable to communicate with the database, the web server should always redirect the shopper to a static page, that indicates:

System Message

*[www.mywebstore.com] is currently unavailable to service your request.
We are sorry for the inconvenience and encourage you to please try again later.*

*[Name of Customer Service Manager],
Customer Service Manager*

get_sales_person_list (OUT fc_sales_person_list)

Returns a list of sales persons (representatives) for the web store.

FC SALES PERSON LIST

Seq.	Column	Data Type	Data Length	Description
1.	Sales_Person_ID	Number		Represents the ID of a sales person.
2.	Sales_Person_Job_Title	String		Represents the job title of a sales person.
3.	Sales_Person_First_Name	String		Represents the first name of the sales person.
4.	Sales_Person_Middle_Initial	String		Represents the middle initial of the sales person.
5.	Sales_Person_Last_Name	String		Represents the last name of the sales person.
6.	Sales_Person_Name_Suffix	String		Represents the name suffix of the sales person.
7.	Sales_Person_Company_Name	String		Represents the company name of the sales person for a multi-company organization.
8.	Sales_Person_Address1	String		Represents the first address line of the sales person's business address.
9.	Sales_Person_Address2	String		Represents the second address line of the sales person's business address.
10.	Sales_Person_City	String		Represents the city of the sales person's business address.
11.	Sales_Person_County	String		Represents the county of the sales person's business address.
12.	Sales_Person_State	String		Represents the state of the sales person's business address.
13.	Sales_Person_Postal_Code	String		Represents the postal code of the sales person's business address.
14.	Sales_Person_Postal_Code4	String		Represents the postal code extension of the sales person's business address.
15.	Sales_Person_Telephone	String		Represents the telephone number of the sales person's business address.

FC SALES PERSON LIST

Seq.	Column	Data Type	Data Length	Description
16.	Sales_Person_Extension	String		Represents the telephone extension of the sales person's business address.
17.	Sales_Person_Mobile_Phone	String		Represents the cellular phone number of the sales person.
18.	Sales_Person_Fax	String		Represents the fax number of the sales person's business address.
19.	Sales_Person_Pager	String		Represents the pager number of the sales person.
20.	Sales_Person_Pager_PIN	String		Represents the pager PIN of the sales person.
21.	Sales_Person_Email	String		Represents the email address of the sales person.
22.	Sales_Person_From_Postal_Code	Number		Represents the starting postal code of the sales person's territory.
23.	Sales_Person_To_Postal_Code	Number		Represents the ending postal code of the sales person's territory.
24.	Sales_Person_Inactive_Flag	Number		Represents the status of the sales person. If flagged then the sales person is inactive.
25.	Sales_Person_Sort_Order	Number		Represents the sort order of the sales person.
26.	Sales_Person_Manager_ID	Number		Represents the ID of the sales person's manager.

Note: NO API functions exist for creating; updating or deleting a Sales Person, since these operations can be performed directly from *inPowerSuite's* Manager application.

Shopping API Functions

get_plugin_list (OUT fc_category_list)

Returns a result set of active Plug-ins available on the web store, using the same ref cursor as defined for **get_category_list()** API function.

get_category_list (IN fn_start_category_id, fn_max_level, OUT fc_category_list)

Returns a tree (hierarchy) of active categories starting at the specified category, up to the max level. If the starting category is zero (0), then all categories are returned up to the max level. If max level is zero (0) then all active categories are returned. The ref cursor contains the following information for categories:

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_start_category_id	Number		Represents the starting category ID.
2.	fn_max_level	Number		Represents the maximum level of categories to return. Level values are: 1 Top level 2 Second level (sub-category) 3 Third level (sub sub-category) 4 Etc.

FC CATEGORY LIST

Seq.	Column	Data Type	Data Length	Description
1.	Category_ID	Number		An inPowerSuite controlled field that is designated as the unique identifier for the Category record.
2.	Category_Name	String		Represents the (full) name of a shopping category. A shopping category defines a grouping mechanism to organize and expose items for a similar manufacturer, brand, product line, genre, collection, etc.

FC CATEGORY LIST

Seq.	Column	Data Type	Data Length	Description
3.	Category_Level	Number	1	Represents the level in the tree (hierarchy) where the category has been defined. Values are: 1 Top level 2 Second level (sub-category) 3 Third level (sub sub-category) 4 Etc.
4.	Parent_Category_ID	Number		Represents another category that you can designate as the parent of the current category record. This relationship allows you to define a category hierarchy and control the drill-down display of categories on your web store.
5.	Sort_Order	Number		Represents the sort sequence for Categories. This sequence value controls the category sort (in ascending order). If blank, then the listing is sorted by category name.
6.	Small_Image	String		Represents a filename reference to a thumbnail (sized) image on the file server for the shopping category.
7.	Large_Image	String		Represents a filename reference to a large (sized) image on the file server for the shopping category.

This result set is sorted by Level (ASC) + Sort Order (ASC) + Category Name (ASC).

get_category_id (IN fv_category_name, OUT fn_category_id)

Returns the ID for the specified category. This API function should be used prior to obtaining more category information from a category name.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fv_category_name	String		Represents the name of the specified category.

OUT - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_category_id	Number		Represents the corresponding ID for the category name.

get_category_product_list (IN fn_category_id, OUT fc_product_list)

Returns a result set of products and/or product SCS groups, for a specified category.
The ref cursor contains the following information.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_category_id	Number		Represents the ID of the specified category.

FC PRODUCT LIST

Seq.	Column	Data Type	Data Length	Description
1.	Product_ID	Number		An inPowerSuite controlled field that is designated as the unique identifier for the product. If the ID were blank, then row in the result set would represent a product SCS group record.
2.	Product_SCS_Group_ID	Number		An inPowerSuite controlled field that is designated as the unique identifier for the product SCS group. If the ID were blank, then row in the result set would represent a product record.
3.	Product_Number	String		Represents the item number that is display on your web store for the selected product or product SCS group. This number can be comprised of any combination of letters and numbers, up to 20 characters in length.
4.	Product_Description	String		Represents a concise description of the product or product SCS group.
5.	Display_Style	String		Represents the drill-down display style for a product SCS group record. The values are E – Exploded D – Dropdown
6.	Product_Style	String		Represents the STYLE value of the Product that does NOT belong to a Product SCS grouping.
7.	Product_Color	String		Represents the COLOR value of the Product that does NOT belong to a Product SCS grouping.

FC PRODUCT LIST

Seq.	Column	Data Type	Data Length	Description
8.	Product_Size	String		Represents the SIZE measurement of the Product that does NOT belong to a Product SCS grouping.
9.	Minimum_Product_Price	Number		Represents the default store price (selling price) for the product, or the minimum price within a product SCS group.
10.	Maximum_Product_Price	Number		Represents the maximum selling price within a product SCS group.
11.	Minimum_Product_Discount	Number		Represents the discount amount (as defined on the active price list) for the product, or the minimum discount within a product SCS group.
12.	Maximum_Product_Discount	Number		Represents the maximum discount amount within a product SCS group.
13.	Minimum_Product_Discounted_Price	Number		Represents the discounted price (as defined on the active price list) for the product, or the minimum discounted price within a product SCS group.
14.	Maximum_Product_Discounted_Price	Number		Represents the maximum discounted price within a product SCS group.
15.	Small_Image	String		Represents a filename reference to a thumbnail (sized) image for the Item.
16.	Sort_Order	Number		Represents the sort sequence for Products. This sequence value controls the category sort (in ascending order). If blank, then the listing is sorted by product description.

The result set is sorted by Sort Order (ASC) + Product Description (ASC).

get_related_product_list (IN fn_product_id, fv_relationship_type, OUT fc_product_list)

Returns a result set of products that are related to the specified product and its relationship type, using the same ref cursor as defined for **get_category_product_list()** API function.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_product_id	Number		Represents the ID of the specified product.
2.	fv_relationship_type	String		Represents a type of product relationship. Values are: Up-Sell and Side-Sell, Substitute, and Kit.

get_product_list (IN search_pattern, search_category, OUT fc_search_product_list)

Returns a result set of products based on the specified Search Pattern and Category. The search is performed across all search-indexed columns in the database. If Category is not specified then all products are searched.

IN – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fv_search_pattern	String		Represents the search value obtained from the shopper/customer for locating products and content on the web store.
2.	fv_search_category	String		Represents the specific shopping category to perform the search.

FC SEARCH PRODUCT LIST

Seq.	Column	Data Type	Data Length	Description
1.	Search_Pattern	String		Represents the search value that was specified by the shopper.
2.	Search_Category	String		Represents the search category that was specified by the shopper.
3.	Number_Of_Hits	Number		Represents the total number for products that match the specified search criteria.
4.	Product_ID	Number		Represents the ID of a product that matches the specified search criteria. If the ID were blank, then row in the result set would represent a product SCS group record.
5.	Product_SCS Group_ID	Number		Represents the ID of a product SCS group that matches the specified search criteria. If the ID were blank, then row in the result set would represent a product record.
6.	Product_Number	String		Represents the Number of a product or the product SCS group that matches the specified search criteria.
7.	Product_Description	String		Represents the Description of a product, or the product SCS group that matches the specified search criteria.
8.	Display_Style	String		Represents the drill-down display style for a product SCS group record. The values are E – Exploded D – Dropdown
9.	Product_Style	String		Represents the STYLE value of the Product that does NOT belong to a Product SCS grouping.
10.	Product_Color	String		Represents the COLOR value of the Product that does NOT belong to a Product SCS grouping.
11.	Product_Size	String		Represents the SIZE measurement of the Product that does NOT belong to a Product SCS grouping.

FC SEARCH PRODUCT LIST

Seq.	Column	Data Type	Data Length	Description
12.	Minimum_Product_Price	Number		Represents the default store price (selling price) for the product or the minimum price within a product SCS group.
13.	Maximum_Product_Price	Number		Represents the maximum selling price within a product SCS group.
14.	Minimum_Product_Discount	Number		Represents the discount amount (as defined on the active price list) for the product, or the minimum discount within a product SCS group.
15.	Maximum_Product_Discount	Number		Represents the maximum discount amount within a product SCS group.
16.	Minimum_Product_Discounted_Price	Number		Represents the discounted price (as defined on the active price list) for the product, or the minimum discounted price within a product SCS group.
17.	Maximum_Product_Discounted_Price	Number		Represents the maximum discounted price within a product SCS group.
18.	Product_Small_Image	String		Reference to the small image file on the web server for the product or the product SCS group that matches the search criteria.

The result set is sorted by Product Description (ASC)

get_exploded_product_info (IN fn_product_id, OUT fc_product_info_list)

Returns a list of products for the specified Product SCS Group. This information can be displayed in any arrangement that is deemed appropriate for your web store, using the same ref cursor as defined in the **get_product_info()** API function.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_product_id	Number		Represents the ID of the specified product SCS group.

get_product_info (IN fn_product_id, OUT fc_product_info_list)

Returns the details for the specified product. This information can be displayed in any arrangement that is deemed appropriate for your web store.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_product_id	Number		Represents the ID of the specified product.

FC PRODUCT INFO LIST

Seq.	Column	Data Type	Data Length	Description
1.	Product_ID	Number		Refer to existing definition.
2.	Product_Number	String		Refer to existing definition.
3.	Product_Description	String		Refer to existing definition.
4.	Product_Price	Number		Refer to existing definition.
5.	Product_Discount	Number		Refer to existing definition.
6.	Product_Discounted_Price	Number		Refer to existing definition.
7.	Product_Additional_Information	String		Represents an additional description of the selected product. Primarily used for Product specification, warranty, limitation and other related information.
8.	Product_Information_URL	String		Represents a fully qualified link to a web page that exposes more content for the selected Product.
9.	Product_Length	Number		An optional field, which represents the LENGTH measurement of the Product.
10.	Product_Width	Number		Represents the WIDTH measurement of the Product.
11.	Product_Height	Number		Represents the HEIGHT measurement of the Product.
12.	Product_LWH_UOM_Code	String		Represents the unit-of-measurement value associated with the length/width/height measurements for the Product.
13.	Product_Weight	Number		A required field, which represents the WEIGHT measurement of the Product that is used for calculating its shipping charge.
14.	Product_Weight_UOM_Code	String		Represents the WEIGHT unit-of-measurement value for the Product
15.	Product_Volume	Number		Represents the VOLUME measurement of the Product that is used for determining the packaging requirements.
16.	Product_Volume_UOM_Code	String		Represents the VOLUME unit-of-measurement value associated with the Product.
17.	Product_Style	String		Represents the STYLE value of the Product that belongs to a Product Style/Color/Size grouping.
18.	Product_Color	String		Represents the COLOR value of the Product that belongs to a Product Style/Color/Size grouping.
19.	Product_Size	String		Represents the SIZE measurement of the Product that belongs to a Product Style/Color/Size grouping.
20.	Marketing_Number	String		Represents the marketing or promotion number that is currently assigned to the product.
21.	Backorder_Lead_Days	Number		Represents the number for days in advance required for reordering stock for the Product, when inventory level is below the threshold.
22.	Manufacturer_Name	String		Represents the name of the Product's manufacturer.
23.	Manufacturer_Part_Number	String		Represents the part number associated with the Product as defined by the manufacturer.

FC PRODUCT INFO LIST

Seq.	Column	Data Type	Data Length	Description
24.	Default Shipping Comment	String		Represents a comment to the shipper regarding the handling and transportation for the Product. For example, the Product's fragility, active ingredients, time sensitivity, etc.
25.	Download_URL	String		Represents the website address to download content for the Product. For example, game software, printer driver, etc.
26.	Default_Ship_Charge	Number		Specifies a default shipping charge for the Product. This defines the overriding shipping charge to use when both the USE DEFAULT SHIP COST FLAG is enabled, and the DEFAULT SHIP CHARGE is greater than store's default shipping charge.
27.	Use_Default_Ship_Cost_Flag	Number		Controls whether the default shipping charge value should be used when calculating the shipping charges for the Product. It defaults to zero (0), thus disabling use of default shipping cost for the Product.
28.	Product_Small_Image	String		Refer to existing definition.
29.	Product_Large_Image	String		Represents a filename reference to a large (sized) image on the file server for the Product.
30.	Product_Style_Image	String		Represents a filename reference to an image on the file server that depicts the Product's style. For example, a fabric swatch for a Pima Cotton Polo Shirt.
31.	Product_Color_Image	String		Represents a filename reference to an image on the file server that depicts the Product's color. For example, a color swatch for a Turquoise Polo Shirt.

Cart API Functions

init_cart (OUT fn_order_id)

Returns a new order ID after initializing an order for the current shopping session.

Note: An *inPowerSuite* order record is initialized with a status of "new" to represent a **cart** and not a true order. Subsequently, the order record then becomes a true order only when the order status is promoted to "booked", thus after the order payment has been authorized.

The calling program should obtain an order ID with this API before calling the **create_order_line()** API function to place a product in the shopping cart.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID for a new order.

add_cart_item (IN fn_order_id, fn_product_id, fn_product_quantity, OUT fn_order_line_id)

This is an overloaded function that returns a new order line ID after adding the specified product to the shopping cart.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID for the specified order.
2.	fn_product_id	Number		Represents the ID of the product to be placed in the cart for the current order.
3.	fn_product_quantity	Number		Represents the quantity for the product being placed in the cart.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_order_line_id	Number		Represents the ID of the new order line for the cart.

add_cart_item (*IN fn_order_id, fv_product_number, fn_product_quantity, OUT fn_order_line_id*)

This is an overloaded function that returns a new order line ID after adding the specified product to the shopping cart.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID for the specified order.
2.	fv_product_number	String		Represents the product number of the product to be placed in the cart for the current order.
3.	fn_product_quantity	Number		Represents the quantity for the product being placed in the cart.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_order_line_id	Number		Represents the ID of the new order line for the cart.

get_cart_items (*IN fn_order_id, OUT fc_cart_item_list*)

Returns a list of products currently in the shopping cart for the specified order.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of the specified order.

FC CART ITEM LIST

Seq.	Column	Data Type	Data Length	Description
1.	Order_ID	Number		Represents the ID of the specified order.
2.	Order_Line_ID	Number		Represents the ID of an order line.
3.	Product_Number	String		Represents the Product's Number for an order line.
4.	Product_Description	String		Represents the Product's Description for an order line.
5.	Product_Quantity	Number		Represents the Product's Quantity for an order line.

FC CART ITEM LIST

Seq.	Column	Data Type	Data Length	Description
6.	Product_Quantity_UOM_Code	String		Represents the Product's unit-of-measurement code for an order line.
7.	Product_Price	Number		Represents the Product's Price for an order line.
8.	Product_Price_Total	Number		Represents the Product's total for an order line, using the formula, Product Quantity times Product Price.
9.	Product_Discount	Number		Represents the Product's discount for an order line.
10.	Product_Discount_Total	Number		Represents the Product's total discount for an order line.
11.	Product_Tax_Rate	Number		Represents the Product's tax rate for an order line.
12.	Product_Selling_Price	Number		Represents the Product's selling price for an order line (less discounts).
13.	Product_Selling_Price_Total	Number		Represents the Product's total selling price for an order line (less discounts).
14.	Ship_Comment	String		Represents a shipping comment provided by the customer (for delivery instructions).

The result set is sorted by Order_ID (ASC) + Order_Line_ID (ASC).

change_cart_item_quantity (IN fn_order_line_id, fn_new_quantity)

Updates the quantity for the specified product in the shopping cart. If the new quantity is zero (0) then the item is removed from the shopping cart.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_line_id	Number		Represents the ID for the order line whose quantity will be changed.
2.	fn_new_quantity	Number		Represents the new replacement quantity for the order line.

remove_cart_item (IN fn_order_line_id)

Removes the specified product from the shopping cart, using the **change_cart_item_quantity()** API setting fn_new_quantity IN parameter to zero (0).

Note: This function will physically delete an order line from a shopping cart and not a "live" order.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_line_id	Number		Represents the ID for the current order line to be removed from the cart.

Checkout API Functions

init_customer (IN fn_customer_id)

Initializes the global private variable gn_cust_id. It is not necessary for the calling program to use this API when creating a new customer, as the **create_customer()** API function makes an internal call to it.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID for the new customer.

add_customer (IN fn_session_id, ..., OUT fn_customer_id)

Returns a new customer ID after creating a new customer with billing information, and calling **init_customer()** API function to set the global private variable gn_customer_id.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_session_id	Number		Represents the ID of the current session.
2.	fv_customer_number	Number		Represents the number assigned to the new customer.
3.	fv_billto_first_name	String		Represents the first name of the new customer.
4.	fv_billto_middle_initial	String		Represents the middle initial of the new customer.
5.	fv_billto_last_name	String		Represents the last name of the new customer.
6.	fv_billto_name_suffix	String		Represents the name suffix of the new customer.
7.	fv_billto_company_name	String		Represents the company name for the new customer.
8.	fv_billto_addres1	String		Represents the first address line for the new customer.
9.	fv_billto_address2	String		Represents the second address line for the new customer.
10.	fv_billto_city	String		Represents the city for the new customer.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
11.	fv_billto_county	String		Represents the county for the new customer.
12.	fv_billto_state	String		Represents the state for the new customer.
13.	fv_billto_postal_code	String		Represents the postal code for the new customer.
14.	fv_billto_postal_code4	String		Represents the postal code extension for the new customer.
15.	fv_billto_country	String		Represents the country for the new customer.
16.	fv_billto_telephone	String		Represents the telephone number of the new customer.
17.	fv_billto_extension	String		Represents the telephone extension of the new customer.
18.	fv_billto_email	String		Represents the email address of the new customer.
19.	fv_sales_person_id	Number		Represents the sales person assignment for the new customer.
20.	fv_billto_username	String		Represents the username of the new customer.
21.	fv_billto_password1	String		Represents the first password entry for the new customer.
22.	fv_billto_password2	String		Represents the second password for the new customer. Note: password1 and password2 entries must be identical.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID of the new customer.

authenticate_customer (IN fv_username, fv_password, OUT fn_customer_id)

Returns an existing customer ID after authenticating the returning customer based on the specified login credentials. It subsequently calls **init_customer()** API function to initialize the global private variable gn_cust_id.

The calling program must use this API function before any customer centric functions can be performed on an existing customer.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fv_username	String		Represents the username for an existing customer.
2.	fv_password	String		Represents the password for an existing customer.

OUT - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID of an existing customer.

verify_customer (IN fv_email_address)

Verify a customer who has forgotten his/her password, based on valid a email address (that exists on file.)

Note: The customer record must be flagged as "active" before the email address is deemed valid.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fv_email_address	String		Represents the password for an existing customer.

This inPowerSuite API function will provide a return code value, as follows:

- 0 – When the email address exists (on file), then each active customer will automatically receive an email containing their username and password information.
- 1 – Email address does not exist on file or no active customer has this email address.

get_customer_billing_info (IN fn_customer_id, OUT fc_customer_billing_info)

Returns the billing information for the specified customer.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID for an existing customer.

FC CUSTOMER BILLING INFO

Seq.	Column	Data Type	Data Length	Description
1.	Customer_ID	Number		An inPowerSuite controlled field that is designated as the unique identifier for the Customer record.
2.	Customer_Number	String		Identifies the customer by a number. This number can be generated by inPowerSuite or obtained from another system. If the former, inPowerSuite will use the customer's telephone number and extension. To ensure uniqueness, inPowerSuite will append a numeric value at the end of the

FC CUSTOMER BILLING INFO

Seq.	Column	Data Type	Data Length	Description
				phone number. For example, 999.999.9999.1
3.	Billto_First_Name	String		Represents the customer's first (or given) name for the billing address.
4.	Billto_Middle_Initial	String		Represents the customer's middle initial for the billing address.
5.	Billto_Last_Name	String		Represents the customer's last (or sur) name for the billing address.
6.	Billto_Name_Suffix	String		Represents the customer's suffix for the billing address. For example, "II" or "Jr.", etc.
7.	Billto_Company_Name	String		Represents the customer's company name for the billing address.
8.	Billto_Address1	String		Represents the first line of the customer's, billing address.
9.	Billto_Address2	String		Represents the second line of the customer's billing address.
10.	Billto_City	String		Represents the city of the customer's billing address.
11.	Billto_County	String		Represents the county of the customer's billing address.
12.	Billto_State	String		Represents the state of the customer's billing address.
13.	Billto_Postal_Code	String		Represents the postal code of the customer's billing address.
14.	Billto_Postal_Code4	String		Represents the postal code extension of the customer's billing address.
15.	Billto_Country	String		Represents the country of the customer's billing address.
16.	Billto_Telephone	String		Represents the customer's billing telephone number. <i>inPowerSuite</i> provides a default format of (999) 999-9999, therefore it is only necessary to enter the digitals (without the format).
17.	Billto_Extension	String		Represents the customer's billing telephone extension.
18.	Billto_Email	String		Represents the customer's billing email address. This address is used for notifying the customer of order confirmations, cancellations, and marketing promotions. Additionally, this address is used for notifying the customer of shipping confirmations when no other email address exists.

get_customer_shipto_list (IN fn_customer_id, OUT fc_customer_shipto_list)

Returns a list of active shipping addresses for the specified customer.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID of the specified customer.

FC CUSTOMER SHIPTO INFO

Seq.	Column	Data Type	Data Length	Description
1.	Customer_ID	Number		Represents the ID of an existing customer.
2.	Shipto_Address_ID	String		Represents the ID of a customer's shipping address.
3.	Shipto_First_Name	String		Represents the customer's first (or given) name for the shipping address.
4.	Shipto_Middle_Initial	String		Represents the customer's middle initial for the shipping address.
5.	Shipto_Last_Name	String		Represents the customer's last (or sur) name for the shipping address.
6.	Shipto_Name_Suffix	String		Represents the customer's suffix for the shipping address. For example, "II" or "Jr.", etc.
7.	Shipto_Company_Name	String		Represents the customer's company name for the shipping address.
8.	Shipto_Address1	String		Represents the first line of the customer's, shipping address.
9.	Shipto_Address2	String		Represents the second line of the customer's shipping address.
10.	Shipto_City	String		Represents the city of the customer's shipping address.
11.	Shipto_County	String		Represents the county of the customer's shipping address.
12.	Shipto_State	String		Represents the state of the customer's shipping address.
13.	Shipto_Postal_Code	String		Represents the postal code of the customer's shipping address.
14.	Shipto_Postal_Code4	String		Represents the postal code extension of the customer's shipping address.
15.	Shipto_Country	String		Represents the country of the customer's shipping address.
16.	Shipto_Telephone	String		Represents the customer's shipping telephone number. inPowerSuite provides a default format of (999) 999-9999, therefore it is only necessary to enter the digitals (without the format).
17.	Shipto_Extension	String		Represents the customer's shipping telephone extension.
18.	Shipto_Email	String		Represents the customer's shipping email address. This address is used for notifying the customer of order confirmations, cancellations, and marketing promotions.

The result set is sorted by Customer_ID (ASC) + Sort_Order (ASC).

add_customer_shipto (IN *fn_customer_id*, *fv_shipto_first_name*,..., OUT *fv_shipto_address_id*)

Returns a new shipping address ID after adding the address for the specified customer.

IN – PARAMTER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID of the specified customer.
2.	fv_shipto_first_name	String		Represents the customer's first (or given) name for the new shipping address.
3.	fv_shipto_middle_initial	String		Represents the customer's middle initial for the new shipping address.
4.	fv_shipto_last_name	String		Represents the customer's last (or sur) name for the new shipping address.
5.	fv_shipto_name_suffix	String		Represents the customer's suffix for the new shipping address. For example, "II" or "Jr.", etc.
6.	fv_shipto_company_name	String		Represents the customer's company name for the new shipping address.
7.	fv_shipto_address1	String		Represents the first line of the customer's, new shipping address.
8.	fv_shipto_address2	String		Represents the second line of the customer's new shipping address.
9.	fv_shipto_city	String		Represents the city of the customer's new shipping address.
10.	fv_shipto_county	String		Represents the county of the customer's new shipping address.
11.	fv_shipto_state	String		Represents the state of the customer's new shipping address.
12.	fv_shipto_Ppstal_code	String		Represents the postal code of the customer's new shipping address.
13.	fv_shipto_postal_code4	String		Represents the postal code extension of the customer's new shipping address.
14.	fv_shipto_country	String		Represents the country of the customer's new shipping address.
15.	fv_shipto_telephone	String		Represents the telephone number for the customer's new shipping address. inPowerSuite provides a default format of (999) 999-9999, therefore it is only necessary to enter the digitals (without the format).
16.	fv_shipto_extension	String		Represents the telephone extension of the

IN – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
				customer's new shipping address.
17.	fv_shipto_email	String		Represents the email address for the customer's new shipping address. This address is used for notifying the customer of order confirmations, cancellations, and marketing promotions.

OUT – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_shipto_address_id	Number		Represents the ID of a new customer's shipping address.

change_customer_shipto (IN fn_customer_id, fv_shipto_address_id,...)

Saves changes for an existing customer's shipping address.

Note: Setting fn_shipto_inactive_flag to one (1) instructs *inPowerSuite* to perform a logical delete of the shipping address, thus disabling it from future use.

IN – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID of the specified customer.
2.	fn_shipto_address_id	Number		Represents the ID of an existing customer's shipping address.
3.	fv_shipto_first_name	String		Represents the customer's first (or given) name for the existing shipping address.
4.	fv_shipto_middle_initial	String		Represents the customer's middle initial for the existing shipping address.
5.	fv_shipto_last_name	String		Represents the customer's last (or sur) name for the existing shipping address.
6.	fv_shipto_name_suffix	String		Represents the customer's suffix for the existing shipping address. For example, "II" or "Jr.", etc.
7.	fv_shipto_company_name	String		Represents the customer's company name for the existing shipping address.
8.	fv_shipto_address1	String		Represents the first address line of the customer's existing shipping address.
9.	fv_shipto_address2	String		Represents the second address line of the customer's existing shipping address.
10.	fv_shipto_city	String		Represents the city of the customer's existing shipping address.

11.	fv_shipto_county	String		Represents the county of the customer's existing shipping address.
12.	fv_shipto_state	String		Represents the state of the customer's existing shipping address.
13.	fv_shipto_Ppstal_code	String		Represents the postal code of the customer's existing shipping address.
14.	fv_shipto_postal_code4	String		Represents the postal code extension of the customer's existing shipping address.
15.	fv_shipto_country	String		Represents the country of the customer's existing shipping address.
16.	fv_shipto_telephone	String		Represents the telephone number for the customer's existing shipping address. inPowerSuite provides a default format of (999) 999-9999, therefore it is only necessary to enter the digitals (without the format).
17.	fv_shipto_extension	String		Represents the telephone extension of the customer's existing shipping address.
	fv_shipto_email	String		Represents the email address for the customer's existing shipping address. This address is used for notifying the customer of order confirmations, cancellations, and marketing promotions.

set_order_line_shipto (IN fn_order_line_id, OUT fn_shipto_address_id)

Sets the shipping destination for the specified order line, as *inPowerSuite* natively supports split shipments at the order line level.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_line_id	Number		Represents the ID of the specified order line.
2.	fn_shipto_address_id	Number		Represents the ID of an existing Customer Shipping Address.

get_UPS_shipping_services (IN fn_order_id, OUT fc_UPS_ship_service_list)

Returns a list of shipping methods and charges for each shipping address that is associated with the order.

This API function should be called after setting the shipping destination for each cart item.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
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IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of the specified order.

FC UPS SHIP SERVICE LIST

Seq.	Column	Data Type	Data Length	Description
1.	Shipto_Address_ID	Number		Represents the ID of a shipping destination associated with the order.
2.	UPS_Ship_Method_Code	String		Represents the code of a valid UPS ship method for the specified shipping destination.
3.	UPS_Ship_Methid_Name	String		Represents the name of a valid UPS ship method for the specified shipping destination.
4.	Ship_Method_Cost	Number		Represents the UPS charge for the ship method and shipping destination.

set_order_shipto_charge (IN fn_order_id, fv_ship_method_code, fn_amount)

Sets the selected shipping method and charge for the specified order.

This API function should be used only when one (1) shipping destination exists for the order. Otherwise, the calling program must use the **set_order_line_ship_charge()** API function instead.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of the specified order.
2.	fv_ship_method_code	String		Represents the code of the selected UPS ship method for the specified order.
3.	fn_amount	Number		Represents the shipping charge for the specified order.

set_order_line_shipto_charge (IN fn_shipto_address_id, fv_ship_method_code, fn_amount)

Sets the selected shipping method and charge for the specified shipping destination on the order.

This API function should be used only when multiple shipping destinations exist for the order. Otherwise, the calling program must use the **set_order_ship_charge()** API function instead.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_shipto_address_id	Number		Represents the ID of the specified shipping destination for an order.
2.	fv_ship_method_code	String		Represents the code of the selected UPS ship method for the specified shipping destination.
3.	fn_amount	Number		Represents the shipping charge for the specified shipping destination.

get_payment_methods (OUT fc_payment_method_list)

Returns a list of payment methods that have been defined for the web store.

FC PAYMENT METHOD LIST

Seq.	Column	Data Type	Data Length	Description
1.	Payment_Method_ID	Number		Represents the ID of a payment method.
2.	Payment_Method_Description	String		Represent the name of the payment method.
3.	Payment_Method_Code	String		Represents the type of a payment method, i.e. VISA, MC, CASH, CHECK, IN, GC.

validate_card (IN fv_card_type, fv_account_number)

Validates a credit card, consisting of the LUHN MOD-10 check, and the validity of the card type.

IN - PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fv_card_type	String		Represents the Payment type chosen for the order. Values are: VISA AMEX MasterCard Discover
2.	fv_account_number	String		Represents the account number for payment when the payment type is a valid credit card.

Note: This API function is only required when the calling program has an intrinsic need to interrogate the card before processing or applying a payment. In some cases, the web store may provide special incentives for using a specific credit card after learning what type of credit card is being used in the e-commerce transaction.

Otherwise, the calling program should only invoke the **authorize_payment()** API function, which manages both validation and authorization of a credit card payment, respectively.

tender_payment (*IN fn_order_id, fv_payment_type, fv_account_number, fv_expiration_date*)

Validates, and authorizes a payment with the credit card processor for the specified order. *inPowerSuite* automatically generates an order confirmation email on successful application of the payment.

IN – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of an existing order.
2.	fv_payment_type	String		Represents the Payment type chosen for the order. Values are: VISA AMEX MasterCard Discover
3.	fv_account_number	String		Represents the account number for payment when the payment type is a valid credit card.
4.	fv_expiration_date	Date Time String		Represents the expiration date of the credit card being used for payment.

Note: This API function calls the **validate_card()** API function before authorizing payment with the web store's credit card processor.

Order Tracking API Functions

get_customer_order_list (IN fn_customer_id, OUT fc_customer_order_list)

Returns a list of orders for the specified customer.

IN – PARAMTER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_customer_id	Number		Represents the ID of the specified customer.

FC CUSTOMER ORDER LIST

Seq.	Column	Data Type	Data Length	Description
1.	Customer_ID	Number		Represents the ID of the specified customer.
2.	Order_ID	Number		Represents the ID of an order for the specified customer.
3.	Order_Number	String		Represents the Order Number for the specified customer.
4.	Order_Status	String		Represents the current status of the specified order. Status values are: N New – Shopping Cart B Booked – Order Payment has been authorized. S Shipped – All items have been shipped. C Cancelled – The entire order has been cancelled
5.	Order_Booked_Date	Date Time String		Represents the date of the current order status.

The result set is sorted by Order Number (DESC).

get_order_info (IN fn_order_id, OUT fc_order_header_info, fc_order_totals)

Returns confirmation information for the specified order. This API is generally needed after processing credit card payment for a new order or for obtaining information for order tracking purposes.

The results from this API function is described in **get_order_header_info()** and **get_order_totals()** API functions.

IN – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of an existing order.

get_order_header_info (IN fn_order_id, OUT fc_order_header_info)

Returns header information for the specified order.

IN – PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of the specified order.

FC ORDER HEADER INFO

Seq.	Column	Data Type	Data Length	Description
1.	Order_ID	Number		Represents the ID of the specified order.
2.	Order_Number	String		Represents the order number for the specified order.
3.	Order_Status	String		Represents the current status of the specified order. Status values are: N New – Shopping Cart B Booked – Order Payment has been authorized. S Shipped – All items have been shipped. C Cancelled – The entire order has been cancelled
4.	Order_Booked_Date	Date Time String		Represents the date of the current order status.
5.	Order_Billto_First_Name	String		Represents the customer's first (or given) name on the billing address, for the specified order.
6.	Order_Billto_Middle_Initial	String		Represents the customer's middle initial on the billing address, for the specified order.
7.	Order_Billto_Last_Name	String		Represents the customer's last (or sur) name on the billing address, for the specified order.
8.	Order_Billto_Name_Suffix	String		Represents the customer's suffix on the billing address, for the specified order.
9.	Order_Billto_Company_Name	String		Represents the customer's company name

FC ORDER HEADER INFO

Seq.	Column	Data Type	Data Length	Description
				on the billing address, for the specified order.
10.	Order_Billto_Address1	String		Represents the first line of the customer's billing address, for the specified order.
11.	Order_Billto_Address2	String		Represents the second line of the customer's billing address, for the specified order.
12.	Order_Billto_City	String		Represents the city of the customer's billing address, for the specified order.
13.	Order_Billto_County	String		Represents the county of the customer's billing address, for the specified order.
14.	Order_Billto_State	String		Represents the state of the customer's billing address, for the specified order.
15.	Order_Billto_Postal_Code	String		Represents the postal code of the customer's billing address, for the specified order.
16.	Order_Billto_Postal_Code4	String		Represents the postal code extension of the customer's billing address, for the specified order.
17.	Order_Billto_Country	String		Represents the country of the customer's billing address, for the specified order.
18.	Order_Billto_Telephone	String		Represents the customer's billing telephone number, for the specified order.
19.	Order_Billto_Extension	String		Represents the customer's billing telephone extension, for the specified order.
20.	Order_Billto_Email	String		Represents the customer's billing email address, for the specified order.

get_order_shipto_info (IN fn_order_id, OUT fc_order_shipto_info)

Returns sub-totals for each ship-to address for the specified order.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of the specified order.

FC ORDER SHIPTO INFO

Seq.	Column	Data Type	Data Length	Description
1.	Order_ID	Number		Represents the ID of the specified order.
2.	Order_SH_Charge_ID	Number		Represents the ID for an Order Shipping Destination record.
3.	Shipto_Subtotal	Number		Represents the derived subtotal for the Order

FC ORDER SHIPTO INFO

Seq.	Column	Data Type	Data Length	Description
				Shipping Destination record.
4.	Shipto_Sales_Tax	Number		Represents the derived sales tax amount for the Order Shipping Destination record.
5.	Shipto_Shipping_Charge	Number		Represents the derived shipping charge for the Order Shipping Destination record.
6.	Shipto_Handling_Charge	Number		Represents the derived handling charge for the Order Shipping Destination record.
7.	Shipto_SH_Tax	Number		Represents the derived tax amount for Shipping and Handling, for the Order Shipping Destination record.
8.	Shipto_Total	Number		Represents the derived total for the Order Shipping Destination record.
9.	Shipto_Ship_Carrier	String		Represents the selected shipping carrier for the Order Shipping Destination record.
	Shipto_Ship_Method	String		Represents the selected ship method for the Order Shipping Destination record.

The result set is sorted by Order_ID (ASC) + Order SH Charge ID (ASC).

get_order_lines (IN fn_order_id, OUT fc_order_line_list)

Returns list of order lines for the specified order including product, quantity, price, sales tax, destination and related charges for the destination.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of the specified order.

FC ORDER LINE LIST

Seq.	Column	Data Type	Data Length	Description
1.	Order_ID	Number		Represents the ID for the specified order.
2.	Order_Line_ID	Number		Represents the ID of an order line.
3.	Product_Number	String		Represents the Product's Number for an order line.
4.	Product_Description	String		Represents the Product's Description for an order line.
5.	Product_Quantity	Number		Represents the Product's Quantity for an order line.
6.	Product_Quantity_UOM_Code	String		Represents the Product's unit-of-measurement code for an order line.
7.	Product_Price	Number		Represents the Product's Price for an order line.
8.	Product_Price_Total	Number		Represents the Product's total for an order line, using the formula, Product Quantity times Product Price.
9.	Product_Discount	Number		Represents the Product's discount for an order line.

FC ORDER LINE LIST

Seq.	Column	Data Type	Data Length	Description
10.	Product_Discount_Total	Number		Represents the Product's total discount for an order line.
11.	Product_Tax_Rate	Number		Represents the Product's tax rate for an order line.
12.	Product_Selling_Price	Number		Represents the Product's selling price for an order line (less discounts).
13.	Product_Selling_Price_Total	Number		Represents the Product's total selling price for an order line (less discounts).
14.	Ship_Quantity	Number		Represents the Product's shipped quantity.
15.	Ship_Date	Date String		Represents the date when the quantity was shipped for the product.
16.	Ship_Comment	String		Represents a shipping comment provided by the customer (for delivery instructions).
17.	Shipto_First_Name	String		Represents the customer's first (or given) name on the shipping address, for an order line.
18.	Shipto_Middle_Initial	String		Represents the customer's middle initial on the shipping address, for the specified order.
19.	Shipto_Last_Name	String		Represents the customer's last (or sur) name on the shipping address, for an order line.
20.	Shipto_Name_Suffix	String		Represents the customer's suffix on the shipping address, for the order line.
21.	Shipto_Company_Name	String		Represents the customer's company name on the billing address, for the order line.
22.	Shipto_Address1	String		Represents the first line of the customer's shipping address, for the specified order.
23.	Shipto_Address2	String		Represents the second line of the customer's shipping address, for the order line.
24.	Shipto_City	String		Represents the city of the customer's billing address, for the specified order.
25.	Shipto_County	String		Represents the county of the customer's shipping address, for the order line.
26.	Shipto_State	String		Represents the state of the customer's shipping address, for the order line.
27.	Shipto_Postal_Code	String		Represents the postal code of the customer's shipping address, for the order line.
28.	Order_Billto_Postal_Code4	String		Represents the postal code extension of the customer's shipping address, for the order line.

The result set is sorted by Order_ID (ASC) + Order_Line_ID (ASC).

get_order_shipto_tracking (fn_order_SH_charge_id, OUT fc_order_shipto_tracking)

Returns list of tracking numbers and possibly, their related URLs to obtain tracking details for the specified order-shipping destination. This API function should be called after using the **get_order_shipto_info()** API function to obtain a list of shipping destinations for an order.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_SH_charge_id	Number		Represents the ID of the specified Order Shipping Destination record.

FC ORDER SHIPTO TRACKING

Seq.	Column	Data Type	Data Length	Description
1.	Order_SH_Charge_ID	Number		Represents the ID of the specified Order Shipping Destination record.
2.	Ship_Carrier_Code	String		Represents the Code for the Shipping Carrier.
3.	Ship_Method	String		Represents the selected method of shipping.
4.	Product_Total_Tax	Number		Represents the total taxes for the products going to the specified shipping destination.
5.	Shipping_Tax_Total	Number		Represents the shipping taxes for the specified shipping destination.
6.	Handling_Tax_Total	Number		Represents the handling taxes for the products going to the specified shipping destination.
7.	Tracking_Number	String		Represents a tracking number assigned by the Shipping Carrier to the Shipping Destination of the order.
8.	Tracking_URL	String		Represents an inPowerSuite generated URL that can be used to obtain Tracking information from the Shipping Carrier's website for the related tracking number.

The result set is sorted by Order_ID (ASC) + Order SH Charge ID (ASC).

get_order_totals (IN fn_order_id, OUT fc_order_totals)

Returns the grand totals (including Sales Tax and Shipping & Handling charges) for the specified order. One row is returned for a valid order ID.

IN - PARAMETER LIST

Seq.	Parameter	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID for the specified order.

FC ORDER TOTALS

Seq.	Column	Data Type	Data Length	Description
1.	Order_ID	Number		Represents the ID of the specified order.
2.	Order_Subtotal	Number		Represents the derived subtotal for the specified order.

FC ORDER TOTALS

Seq.	Column	Data Type	Data Length	Description
3.	Order_Sales_Tax	Number		Represents the derived sales tax amount for the specified order.
4.	Order_Shipping_Charge	Number		Represents the derived shipping charges for the specified order.
5.	Order_Handling_Charge	Number		Represents the derived handling charges for the specified order.
6.	Order_SH_Tax	Number		Represents the derived tax amount for Shipping and Handling, for the specified order.
7.	Order_Total	Number		Represents the derived grand total for the specified order.

get_order_payment_list (IN fn_order_id, OUT fc_order_payment_list)

Returns payment information for the specified order. This includes all payment adjustments made to the specified order, in event of product returns and/or cancellations.

IN – PARAMETER LIST

Seq.	Column	Data Type	Data Length	Description
1.	fn_order_id	Number		Represents the ID of an existing order.

FC ORDER PAYMENT LIST

Seq.	Column	Data Type	Data Length	Description
1.	Order_ID	Number		Represents the ID for the specified order.
2.	Order_Payment_ID	Number		Represents the ID for payment record associated with the specified order.
3.	Exchange_Rate	Number		Represents the exchange rate associated with the order payment record.
4.	Payment_Amount	Number		Represents the payment amount associated with the order payment record.
5.	Credit_Card_Name	String		Represents the credit card holder identified on the order payment record.
6.	Credit_Card_Number	String		Represents the credit card number identified on the order payment record.
7.	CC_Expiration_Date	Date Time String		Represents the expiration date of the credit card identified on the order payment record.

Examples

Displaying a completed order for confirmation purpose.

As you may have realized, information for an order is very complex and is comprised of literally dozens of distinct pieces of information. By design, *inPowerSuite* manages the complexity of an order by breaking it into discrete atomic parts.

inPowerSuite could have provided a single API function that a calling program can use to obtain all details for a specific order, in one stab. However, the result set of such an API function would literally contain over one hundred fields of information for the order, thus becoming too unwieldy for any calling program to manage, before attempting to present it to the customer.

Instead, *inPowerSuite* provides API functions that would obtain the order in its distinct atomic parts, which the calling program can assemble into any desired presentation for the customer.

The hierarchical outline (tree) below, illustrates the various atomic parts used by *inPowerSuite* to manage a customer's order.

Seq.	Order Section	Description
1.	Order	Represents a specific order header.
2.	Billing Information	Represents billing information for the specified order.
3.	Order Summary	Represents subtotal, sales tax, shipping and handling charges, shipping and handling taxes, and order grand total, for the specified order.
4.	Order Lines	Represents product, quantity, selling price, product total, sales tax, shipping and handling charges, and the shipping address identifier per order line for the specified order.
5.	Shipping Destinations	Represents the shipping address information for each destination associated with the order.
6.	Tracking Numbers	Represents the tracking numbers obtained from the shipping carrier for the each destination specified for an order line.
7.	Order Payments	Represents payment information for the specified order including payment adjustments.

Since *inPowerSuite* provides specific APIs to obtain these discrete atomic sections of order information, you can use the API mapping below as a reference to access all information pertinent to a customer's order.

Seq.	Order Section	API Call
1.	Order	Use <code>get_customer_order_list()</code> to obtain a list of orders.
2.	Billing Information	Use <code>get_order_info()</code> for a specific order.

Seq.	Order Section	API Call
3.	Order Summary	Use <code>get_order_totals()</code> for a specific order.
4.	Order Lines	Use <code>get_cart_items()</code> for a specific order.
5.	Shipping Destinations	Use <code>get_shipto_address_list()</code> for a specific order.
6.	Tracking Numbers	Use <code>get_order_tracking_list()</code> for a specific order.
7.	Order Payments	Use <code>get_order_payment_list()</code> for a specific order.

PL/SQL calling program

From PL/SQL, a developer can create a wrapper procedure that calls a specific *inPowerSuite* API function.

Outlined below are the required steps for building the wrapper procedure for an explicit *inPowerSuite* API function, and executing it from a SQLPlus prompt.

There are two distinct examples provided for your benefit. The first example illustrates how to use the **get_category_product_list()** API function to obtain a list of products for a specific category, while the second example illustrates how to use the **get_product_info()** ASP function to obtain details for a specific product.

Obtain a list of products for a specific category:

*First, define an explicit wrapper PL/SQL procedure that will be used to invoke the `get_category_product_list` *inPowerSuite* API function.*

```

procedure wrapper_get_category_product_list(
    fn_category_id in number,
    fn_error_code out number,
    fc_product_list out ngsapi.c_type)
IS
BEGIN
    fn_error_code := NGSAPI.get_category_product_list(fn_category_id,
    fc_product_list);
    if ln_return <> 0 then
        call wrapper_error_handling()
        exit;
    end if;
END;
```

*Secondly, use SQLPlus program to test the *inPowerSuite* API function, you must establish a connection to the *inPowerSuite* database (where the NGSAPI package exists). Subsequently, declare a variable for API result set and another for the API returncode. Additionally, you should set the autoprint and serveroutput environment variables for your SQLPlus session.*

Finally, execute the wrapper PL/SQL procedure.

```

SQL> variable c refcursor;
SQL> variable e number;
SQL> set autoprint on;
SQL> set serveroutput on;
SQL> exec test_get_category_product_list(1014,:e,:c)
```

Below are the results from the API's ref cursor.

CATEGORY_ID	PRODUCT_ID	PRODUCT_NUMBER	PRODUCT_SCS_GROUP_ID	DISPLAY_ST	PRODUCT_DESCRIPTION
PRODUCT_SIZE	SORT_ORD				
	1014	0	1001	EX	Niche Polo Shirt
				1	
	1014	0	1000	EX	Niche Uniball Pen
				1	
	1014	1036 5205	0		Niche Dry Erase Markers
				2	
	1014	1037 5210	0		Niche Highlighters
				2	
M	1014	1030 5010	0		Niche Long Sleeve Denim
				2	
L	1014	1031 5012	0		Niche Long Sleeve Denim
				2	
	1014	1032 5014	0		Niche Long Sleeve Denim
				2	
XL					
	1014	1027 5005	0		Niche Tee Shirt
M				2	
	1014	1028 5006	0		Niche Tee Shirt
L				2	
	1014	1029 5007	0		Niche Tee Shirt
XL				2	

10 rows selected.

Obtain product information for a specific product:

First, define an explicit wrapper PL/SQL procedure that will be used to invoke the `get_product_info` inPowerSuite API function.

```
CREATE OR REPLACE
Package      NGSAPI_CALLING_EXAMPLE
IS
/*****
*          NGSAPI_CALLING_EXAMPLE
*****/
```

type c_type is ref cursor; - Declare a ref cursor in the calling program wrapper function

```
type gr_product_info_rec is RECORD
    (product_id          number,
     product_number      varchar2(20),
     product_description  varchar2(80),
     product_price       number,
     product_discount    number,
     product_discount_price number,
     product_additional_info varchar2(2000),
     product_info_url    varchar2(200),
     product_length      number,
     product_width       number,
     product_height      number,
     product_lwh_uom_code varchar2(12),
     product_weight      number,
     weight_uom_code     varchar2(12),
     product_volume      number,
     volume_uom_code     varchar2(12),
     product_style       varchar2(60),
     product_color       varchar2(60),
     product_size        varchar2(60),
     marketing_number    varchar2(20),
     backorder_lead_days number,
```

```

        manufacturer_name      varchar2(60),
        manufacturer_part_number varchar2(20),
        default_shipper_comment varchar2(2000),
        download_url            varchar2(2000),
        default_ship_charge     number(11,2),
        use_default_ship_cost_flag number(1,0),
        image_filename_small     varchar2(100),
        image_filename_large     varchar2(100),
        image_filename_style     varchar2(100),
        image_filename_color     varchar2(100));

type gr_category_list_rec is RECORD
    (category_level      number,
     category_id         number,
     category_name       varchar2(60),
     parent_category_id  number,
     image_filename_small varchar2(100),
     image_filename_large varchar2(100),
     sort_order          number(6,0));

CREATE OR REPLACE
Package Body      NGSAPI_CALLING_EXAMPLE
IS

PROCEDURE wrapper_get_product_info
(   fn_product_id      IN number)
IS
    ln_return number;          -- Return code variable
    lv_out varchar2(1000);     --
    lc_out c_type;            --
    lc_row gr_product_info_rec; --

BEGIN
    ln_return := ngsapi.get_product_info(fn_product_id, lc_out);

    if ln_return <> 0 then
        call wrapper_error_handling()
        exit;
    end if;

LOOP
    fetch lc_out into lc_row;
    exit when lc_out%notfound;

    lv_out :=          'product_id: '      || lc_row.product_id ||
                      'product_number: '  || lc_row.product_number ||
                      'product_description: ' ||
                      'product_price: '    || lc_row.product_price;

    dbms_output.put_line (lv_out);
end loop;

END get_product_info;

END; -- Package Body NGSAPI_CALLING_EXAMPLE

```

Secondly, use SQLPlus program to test the inPowerSuite API function, you must establish a connection to the inPowerSuite database (where the NGSAPI package exists). Subsequently, declare a variable for API result set and another for the API returncode. Additionally, you should set the autoprint and serveroutput environment variables for your SQLPlus session. Finally, execute the wrapper PL/SQL procedure.

```

SQL> set serveroutput on
SQL> exec ngsapi_calling_example.get_product_info(1022);
SQL> product_id: 1022 product_number: 3502 product_description: product_price: 0
    PL/SQL procedure successfully completed.
SQL>

```

Active Server Page (ASP) calling program

Obtain a list of products for a specific category.
Obtain product information for a specific product.

ColdFusion calling program

Obtain a list of products for a specific category.
Obtain product information for a specific product.

C++ calling program

Obtain a list of products for a specific category.
Obtain product information for a specific product.

Notes